

Human Detection Based Yolo Backbones-Transformer in UAVs

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Abstract— This study presents a new method for human detection in UAVs using Yolo backbones transformer. The proposed framework utilizes backbones YoloV8s, SC3T (Based Transformer), with RGB inputs to accurately perceive human detection. Experimental results demonstrate that the proposed method achieves an average accuracy of around 90.0% mAP@0.5 for human detection in the Human UAVs dataset, surpassing the performance of competitive baselines. The superior performance of our Deep Neural Network (DNN) can provide context awareness to UAVs. Furthermore, the proposed method can be easily adapted to detect UAVs in various applications. This work highlights the potential of the Yolo backbones transformer for enhancing human detection in UAVs, demonstrating its superiority over conventional methods. Overall, the proposed framework can pave the way for future research in UAV detection applications. Training code and self-collected Human detection dataset are released in <https://github.com/Tyler-Do/Yolov8-Transformer>.

Keywords—Transformer, UAVs, Yolo, human detection, DNN, CNN

I. INTRODUCTION

Unmanned Aerial Vehicles (UAVs), also known as drones or unmanned aircraft, are a type of vehicle that can operate without a human pilot on board. Equipped with cameras, sensors, communication devices, and other payload components [1], UAVs were initially developed for military purposes, but have found increasing use in civilian applications such as border protection, agriculture, and aerial photography [2]. The development of unmanned aircraft systems began with the United States Department of Defense, and leading UAV manufacturing countries currently include the USA, Israel, China, Iran, and Russia [3]. Compared to traditional aircraft, UAVs are simpler and less expensive to manufacture, using essential components.

Detecting humans is an essential task for UAVs across various applications, such as crowd monitoring, identification of potential threats, and ensuring safety. With the aid of

suitable software and sensors, UAVs can detect human presence in real-time, enabling them to aid in search and rescue operations, surveillance, law enforcement, and other scenarios where human activity needs to be monitored from the air [1]. The ability of UAVs to detect humans provides valuable situational awareness, which can help prevent accidents, respond to emergencies, and safeguard human lives. Moreover, Human detection using UAVs can be achieved through various approaches, depending on the type of sensors and algorithms employed. Common sensors utilized for human detection on UAVs include cameras, thermal imaging cameras, Light Detection and Ranging (LIDAR) sensors, and other specialized sensors. These sensors capture either visual or thermal signatures of humans, which are then processed through computer vision and machine learning algorithms to detect and track humans in real time. The success of human detection using UAVs is dependent on the quality and accuracy of the sensors and algorithms used, and researchers continue to explore new and innovative ways to improve the performance of these systems [4], [5].

Transformers have become the dominant model in natural language processing due to their ability to learn complex dependencies between input sequences through self-attention [6]. Recently, transformers have been applied to image processing tasks, specifically object detection, with promising results. While originally developed for natural language processing, transformers are well-suited for processing images as sequences of patches due to their ability to capture complex dependencies between input sequences through self-attention. This enables transformers to gather and associate information from a large neighborhood, improving semantic discrimination and reducing category confusion in complex and large-scale scenes. Furthermore, multi-head self-attention allows transformers to capture global context information, a capability not possible with traditional convolutional neural networks (CNNs) due to the locality of the convolution operation. By retaining spatial information for object detection, transformers have shown the potential to improve the performance of image processing tasks, such as object detection in drone images. [7].

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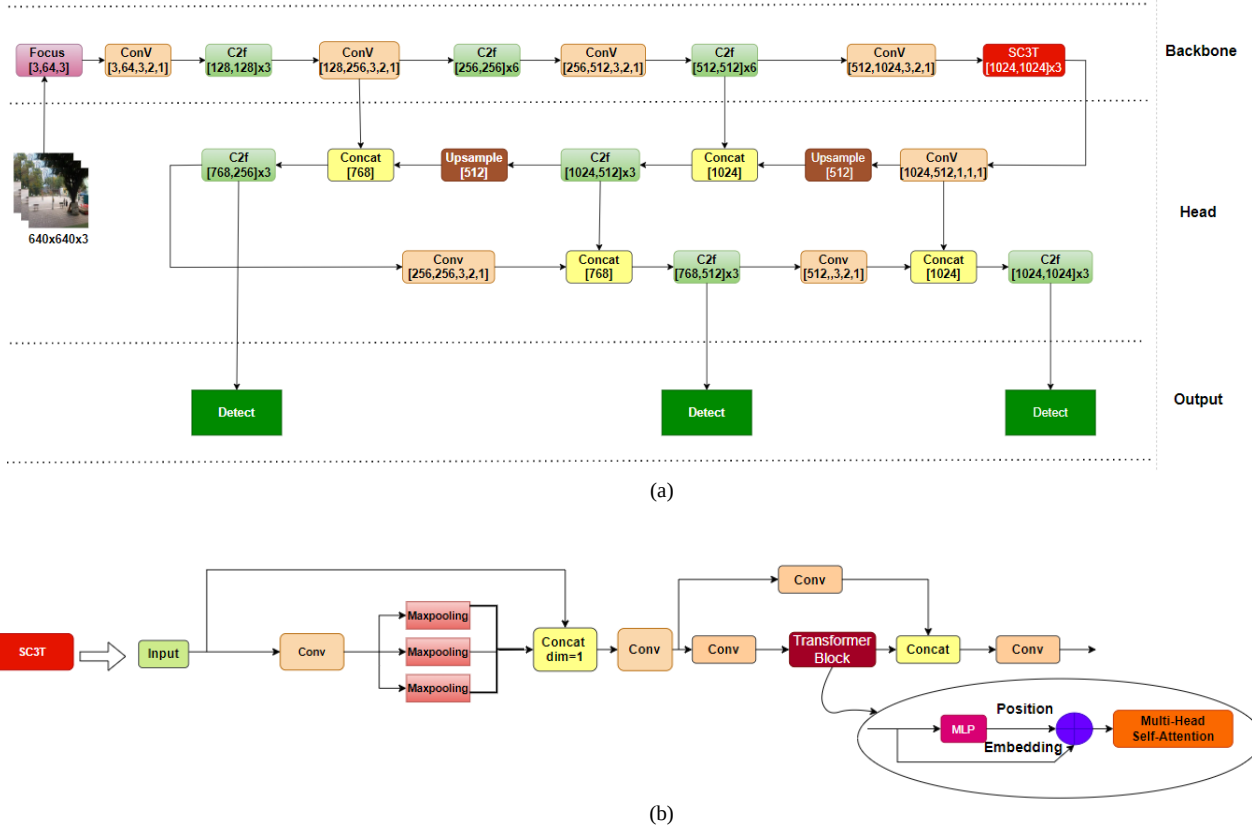


Fig. 1. The proposed YoloV8-Transformer for Human UAVs dataset: (a) the overall architecture, (b) the proposed SC3T.

The effectiveness and efficiency of human detection utilizing YOLO or any object detection algorithm may be influenced by several factors, such as the implementation, the dataset used for training, the computational resources available, and the prevailing environmental conditions.

Consistent research and progress in the areas of computer vision and machine learning techniques may help to alleviate some of these restrictions and enhance the precision and dependability of human detection through cameras mounted on UAVs. One popular approach for human detection is using Deep Neural Network (DNN) algorithms, such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) [5][8][9]. These algorithms can be trained on large datasets of human images, enabling them to learn features and patterns associated with human appearance, motion, and behavior [6]. Once trained, these algorithms can be deployed on UAVs to process the sensor data in real time and detect humans with high accuracy. YOLOv8 [10], on the other hand, was released in January 2023 by Ultralytics, the company that developed YOLOv5 [11]. Since there is currently no published scientific paper on YOLOv8, it is an anchor-free object detection algorithm that reduces the number of box predictions and speeds up the Non-Maximum Suppression process. YOLOv8 also utilizes mosaic augmentation during training, but this technique is disabled for the final ten epochs to avoid any potential negative impacts on performance. Human detection based on YOLO backbones with transformer models is an active area of research. In recent years, due to their potential for accurate and efficient object detection in images and videos. The combination of YOLO backbones and transformer models for human detection has garnered considerable interest owing to their ability to

deliver precise and efficient object detection in images and videos. YOLO backbones that use convolutional neural networks (CNNs) as backbones are known for their speed and accuracy in object detection tasks, including human detection. They typically consist of multiple convolutional layers that extract features from images at different scales for end-to-end detection [12].

In the paper, we mainly adopt a YOLOv8s architecture with modifications that incorporate the SC3T module based on transformers, a popular architecture for sequence prediction. Our proposed DNN predicts the location of all humans from UAVs camera that dataset was created by recording videos using drones and mobile phones in diverse environments, such as parks, houses, and football fields. The images of people were captured under rigorous standards to include a variety of scenarios. Subsequently, we evaluate the proposed DNN on the Human UAVs dataset, against a very competitive YOLOv5 and YOLOv8. They have undergone many design iterations and their performance was greatly improved since the original publication. Our experiments show that our new model achieves comparable performances.

II. PROPOSED YOLOV8S-TRANSFORMER

The proposed architecture enhances the YOLOv8s architecture by incorporating the SC3T module into the final layer, resulting in more effective detection with a smaller network structure as shown in Figure 1. To achieve optimal feature extraction, the backbone of the proposed architecture consists of the focus module, Conv module, C2f, and SC3T module (a combination of C3TR and SPP module). The Conv module uses convolutional layers, batch normalization layers, and the SiLU activation function to extract features. In the

proposed architecture, the first layer's Conv module is replaced with the Focus module, which performs a slice operation on images and concatenates them to improve feature extraction during down-sampling. To address non-uniform input image sizes and fuse features of different sizes, we use the SPP module at the bottom of the backbone. We implement SPP using kernels of uniform stride (stride = 1) but different sizes (5×5 , 9×9 , and 13×13), and feature fusion is achieved by channel stitching (concatenation). Although SPP is traditionally used in the field of natural language processing, our implementation demonstrates its effectiveness in improving object detection in images as shown in Fig. 1(a).

The YOLOv5 architecture incorporates the C3 module, which is composed of three standard convolutional layers and multiple bottleneck modules. The C3 module includes two branches, with one branch utilizing multiple bottleneck stacks and three standard convolution layers, while the other branch passes through only one basic convolution layer before concatenating the two branches. The use of bottleneck modules effectively reduces the number of training parameters and computation, addressing the issues of gradient explosion and gradient disappearance in deep networks, and enhancing the model's learning ability. Moreover, YOLOv8 introduces the C2f module, inspired by the C3 module to obtain more abundant gradient flow information while maintaining a lightweight structure. The C2f module includes a fusion module that fuses features from different layers to improve the gradient flow and enhance the model's representational power. The combination of these advancements results in more accurate and efficient object detection performance.

Fig. 1(b), the SC3T module is an extension of the YOLOv8 architecture that incorporates Transformer blocks at the network's three output detection headers using Concat with weights. The standard Transformer layer is employed to obtain global information from the last block of the Backbone. The Transformer Encoder is composed of a Multi-Head Self-Attention Mechanism, which updates and combines query value, key-value, and weight value containing global feature information from different subspaces for linear projection. The self-attention mechanism effectively captures contextual information and reduces the loss of global information. Finally, the features are non-linearly outputted through MLP to enhance the expression of the self-attentive mechanism. This incorporation of the Transformer block allows for improved performance and accuracy in object detection tasks.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experiment Environment

We trained the proposed YOLOv8s-Transformer model on the Human dataset described below using Google Colab with High RAM Colab Runtime and Tesla V100 GPU. Following the completion of the training process, we obtained corresponding sets of weights for each model. Next, we evaluated the effectiveness of each model based on the test dataset. Finally, we compared the results obtained with those of the original YOLOv5s and YOLOv5s-Transformer versions.

B. Human UAVs dataset

To address the challenge of detecting people from unmanned aerial vehicles (UAVs), we developed a real-world dataset to be used as input for data training. The dataset was constructed by capturing videos from a camera mounted on

drone devices and transmitting the footage to a ground station through the TS832-RC832 wireless transmitter in various scenarios, including parks, houses, and football fields, while images of people were taken under strict criteria to cover a range of different scenarios as shown in Fig. 2. The dataset was created based on three criteria: Scene, Altitude, and Lighting Conditions. For the Scene, we captured images in wooded areas, crowded places, and areas with few people to ensure that the dataset is suitable for search and rescue operations and to prevent obstructions under tree canopies. For the Altitude, we captured images from three different tilt angles: horizontal, low altitude, and straight down with distances ranging from close to far. Lastly, for the Lighting Conditions, we captured videos in sufficient lighting conditions (daytime, noon) and under low light conditions during dusk and night-time with dim lighting. The addition of low-light condition data can assist in search and rescue operations by enabling the drone to search in challenging lighting conditions. After the videos were captured, we used the ffmpeg tool to extract images at a rate of 5 frames per second, resulting in a total of 7,099 images. Following the initial dataset criteria, we removed images that did not meet the criteria or were blurry and noisy, resulting in a final dataset of 2,227 images.



Fig. 2. Example of Human UAVs dataset

To develop an object detection application, we needed to accurately define the bounding boxes containing people in the entire image. Therefore, we utilized the Roboflow tool to create bounding boxes around the "Person" object. After the images were marked, the application returned the corresponding image and label saved in a .txt file. The dataset consisted of 2,227 images with a total of 6,668 bounding boxes. To ensure an adequate training dataset, we divided the dataset into three sets: training, validation, and testing, using an 80:10:10 ratio.

C. Metrics for performance evaluation

To assess the performance of the YOLOv8s-Transformer model for people detection, several evaluation indicators are utilized, including GFLOPS (Giga Floating-point Operations Per Second), precision, recall, and mean average precision (mAP).

GFLOPS, which stands for Giga Floating-point Operations Per Second, is commonly used as an indicator of GPU performance. It represents the number of billion floating-point operations that can be performed per second. During the process of improving a model, it is often the case that GFLOPs and parameters increase. However, it is generally desirable for both of these values to be kept as low as possible.

The precision rate can be computed as the ratio of the number of correctly predicted positive samples by the model to the total number of positive samples predicted:

$$precision = \frac{TP}{TP+FP} \quad (1)$$

The recall rate is a measure of the proportion of actual positive samples that are correctly identified by the model. It is calculated as the ratio of the number of true positive samples to the sum of true positive and false negative samples. The mathematical expression for the recall rate is given by the Equation :

$$recall = \frac{TP}{TP+FN} \quad (2)$$

The precision-recall curve is a graphical representation where the y-axis represents precision and the x-axis represents recall. The region enclosed by the curve and the axes are defined as the average precision (AP). The precision-recall curve displays the accuracy values when more outer boxes are accepted (i.e., lower class probability thresholds leading to higher recall values).

$$AP = \int_0^1 P(R)dR \quad (3)$$

As illustrated below, the mean Average Precision (mAP) is the average precision rate calculated over all classes. The mAP value reflects the overall accuracy of the model, and a higher mAP value indicates a better detection performance and higher accuracy.

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \quad (4)$$

D. Experimental Results and Discussion

In the performance, the three types of DNN have been adopted in the Human UAVs dataset. Type1 is called the Yolov5s base model, Type2 is called Yolov5s-Transformer, Type3 is called the Yolov8s base model and Type4 is called Yolov8s-Transformer. The recall and precision are both crucial for effective human detection in comparison of three types as shown in Fig. 3. A model Yolov8s Transformer with high recall and precision may identify many positive samples. The improved Yolov8s-Transformer, aims to enhance both recall and precision by incorporating mechanisms that can capture global features and increase the receptive field.

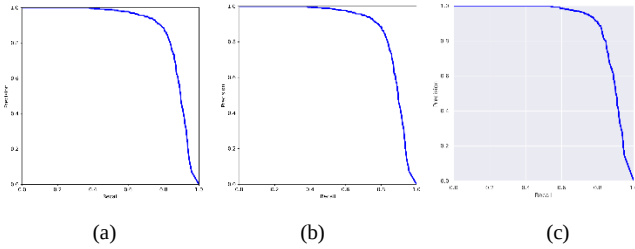


Fig.3. Illustration the curve performance of precision-recall: (a) YoloV5, (b) YoloV5-Transformer, and (c) YoloV8s_Transformer.

Fig. 4(a) and Fig. 4(b) represent the model can locate the center of an object and the predicted bounding box covers an object as well. The Fig. 4(c) and Fig. 4(d) show the training and validation in object loss. This metric measures the difference between the predicted class probabilities and the ground truth class probabilities during training. Fig. 4(g) calculates the mean average precision of the model detection with a minimum overlap of 0.5 between predicted and ground truth bounding boxes. Fig. 4(f) calculates the mean average precision of the model detection with an overlap ranging from

0.5 to 0.95 between predicted and ground truth bounding boxes.

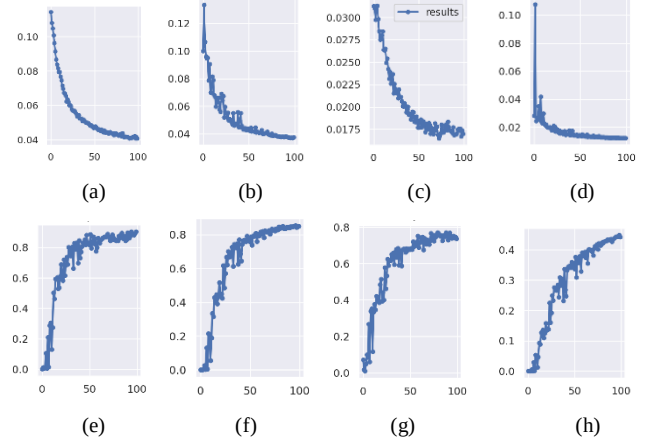


Fig.4. Illustration the metric curve of YoloV8-Transformer: (a) the training box loss, (b) the validated box loss, (c) the training object loss, (d) the validated object loss, (e) the precision, (f) the recall, (g) mAP@0.5, (h) mAP@0.5:0.95.

In Table 1, the comparison of results and performance evaluation of the Yolov5s base model, Yolov5s-Transformer, and Yolov8s-Transformer using 5 metrics including Precision, Recall, mAP@0.5, number of Parameters, and GFLOPS, the accuracy of the base Yolov5s model is 84.6%. After improvement with the Transformer network, it increased to 87.6%. With the proposed Yolov8s-Transformer model, all evaluation metrics are higher compared to the base Yolov8s model, and the mAP@0.5 of the proposed model increased to 90.0%. Additionally, the Recall metric (sensitivity of the model) of the Yolo-Transformer models was consistently above 80.3%, significantly higher than the base model (74.6%) by a margin of 6%. The Precision metric (reliability of the model) of the Yolov8-Transformer model had the highest rate at 90.1%. When combining models to improve accuracy, the model typically becomes more complex, slower in processing speed, larger in size, and requires more computation. In this case, the number of parameters and GFLOPS of the new model increased, but not significantly compared to the base model, and the accuracy increased significantly, indicating a clear improvement in model performance.

Table 1. Experimental of three type model for testing in Human UAVs dataset

Model	Precision (%)	Recall (%)	mAP@0.5 (%)	No. Paramester (Million)	GPL OPS
Type1	89.2	74.6	84.6	7M	15.8
Type 2	88.3	80.3	87.6	7.4M	16.4
Type 3	89.9	79.8	88.9	10.2M	28.8
Type 4	90.1	80.3	90.0	10.5M	29.2

Based on the quantitative results above, we have seen the effectiveness of the Yolov8s-Transformer. To provide a more visual comparison, we compared the detection results of humans in images. As shown in Fig. 5, the first row shows the result of the Yolov5s-Transformer model, and the second row shows the result of the Yolov8-Transformer model. The YOLOv8 model can help in detecting small objects by introducing the Focus, SPP, and Transformer (C3TR)

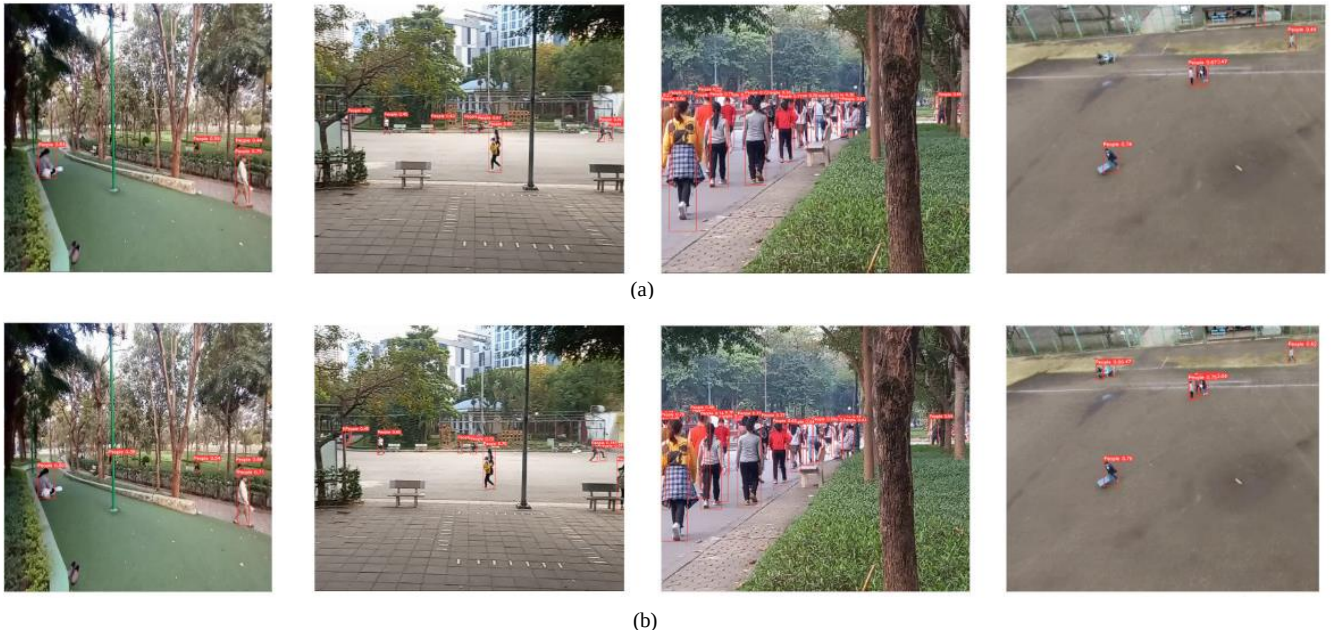


Fig. 5. The visualization of human detection of the Human UAVs dataset where red boxes being the being the predicted boundary associate with labels and their confident scores.(a) Yolov5s-Transformer, (b) Yolov8s-Transformer

mechanisms into the network. The Focus mechanism allows the model to focus on small objects, while the SPP mechanism enhances the network's ability to extract features from the backbone. The Transformer mechanism increases the receptive field of the model, enabling it to capture more global features. The proposed Transformer model has shown excellent results in detecting small-sized objects from far to near perspectives. While the original Yolov5s model still has some missed objects, the proposed model has performed very well in addressing these issues.

IV. CONCLUSION

In this paper, the proposed DNN has successfully to detection the human from UAVs. The primary focus of our study was to develop an improved YOLOv8s model capable of detecting humans in UAV images of various resolutions from the dataset. To achieve this goal, we made three significant modifications to the YOLOv8s backbone, which involved integrating the Focus, SC3T mechanisms into the YOLOv8s network to capture more global features and increase the receptive field. The experimental of four different YOLO models were employed in our study, namely YOLOv5s, YOLOv8s, an enhanced version of YOLOv5s, and an enhanced version of YOLOv8s. The results showed that the proposed improved YOLOv8s-Transformer model achieved the highest precision rate of 90.1%, a recall rate of 80.3%, and a GFLOPS of 29.2. Furthermore, the model achieved a 90.0% mAP, outperforming some state-of-the-art algorithms on the established dataset. To further validate the effectiveness of the proposed algorithm, we conducted empirical analysis on our collected dataset and compared it with the established dataset. Therefore, the proposed YOLOv8s model can serve as an outstanding engine for human localization and classification awareness of human detection on UAVs. In the future, we will continue to improve the network and design the hardware in combination with related topics, providing possible opportunities for the application of the YOLOv8s model in various scenarios, including detecting people in a range of environments.

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